

METHODOLOGY, FORMALIZATION, AND NUMERICAL EVIDENCE (TECHNICAL / COMPUTATIONAL SECTION — DRAFT)

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ABSTRACT. Placeholder. The abstract for the full paper will be drafted jointly once the technical section is finalised.

1. METHODOLOGY, FORMALIZATION, AND NUMERICAL EVIDENCE

The section number X and the equation / theorem / lemma tags (Lemma X.3.1, Theorem X.4.1, etc.) are placeholders to be assigned on integration into the full paper.

This section records the technical and computational content of the paper: the algebraic identities at the heart of the corrected B_∞ framework (§X.3–§X.4), the open challenges that organise the program forward (§X.4.4, §X.7), the numerical evidence at the two scales on which our verification operates (§X.5), and the Lean 4 formalisation inventory (§X.6).

The numerical evidence has **two distinct verified scales**, which we keep rigorously separate throughout. The **replication scale** is $x = 1.3 \cdot 10^{13}$, at which two independent prime-counting implementations agree on Koyama’s Dominance-of- -1 residue tables (§X.5.1). The **analytic-identity scale** is $K \leq 10^8$, at which the corrected B_∞ identity (§X.5.4), the subleading constant C_1 (§X.5.2), and the Aoki–Koyama–Mertens drift toward $e^{-\gamma}$ (§X.5.3, at $K \leq 10^7$) are verified across three numerical stacks. Numbers from one scale are not transferred to the other.

1.1. Notation. Fix a primitive non-principal Dirichlet character χ modulo $q \geq 2$, and let $\rho = \frac{1}{2} + i\tau$ with $\tau \neq 0$ be a simple non-trivial zero of $L(s, \chi)$. We use:

- $E_K(\chi, \rho) := \prod_{p \leq K} (1 - \chi(p) p^{-\rho})^{-1}$ (truncated partial Euler product);
- $c_K(\chi, \rho) := \sum_{n \leq K} \mu(n) \chi(n) n^{-\rho}$ (truncated Möbius sum);
- $D_K(\chi, \rho) := c_K(\chi, \rho) E_K(\chi, \rho)$ (Dirichlet D_K statistic);
- $T_K(\chi, \rho) := \sum_{p \leq K} \sum_{k \geq 2} \chi(p)^k / (k p^{k\rho})$ and its limit T_∞ with $B_\infty := \exp(T_\infty)$;
- $C_1(\chi, \rho) := -L''(\rho, \chi) / (2 L'(\rho, \chi)^2)$ (subleading constant);
- ψ for the primitive character of conductor $f \mid q$ inducing χ^2 .

The branch of $\log L(2\rho, \psi)$ is fixed by analytic continuation from $\operatorname{Re}(s) > 1$, where the absolutely convergent log-Euler expansion applies, to the boundary line $\operatorname{Re}(s) = 1$ via Hadamard–de la Vallée Poussin non-vanishing.

1.2. Methodology of double verification. Every numerical claim of §X.5 is computed by **two independent implementations in two languages with independent algorithms**.

- **L1 (primary).** `mpmath` 1.4 (Python 3.13), 50 decimal places. Direct partial-Möbius and partial-Euler evaluation; each zero ρ refined by Muller’s method to $|L(\rho, \chi)| < 10^{-50}$.
- **L1b (in-language cross-check).** Same library, independent algorithm: Hurwitz-zeta expansion $L(s, \chi) = q^{-s} \sum_{a=1}^q \chi(a) \zeta(s, a/q)$ in place of `mpmath.dirichlet`, central-difference numerical derivatives at three step sizes, and an independent linear sieve for $\mu(n)$.
- **L2 (cross-language).** PARI/GP 2.17.3 (C), 57 dps default; with `python-flint` 0.8.0 / Arb (FLINT 3.3) at 250 bits as a third stack on the worst-case pair.

Acceptance gates: agreement to ≥ 12 significant digits on ρ and to ≥ 8 digits on L' , L'' , C_1 ; for residual quantities where cancellation occurs (the B_∞ residual), agreement to 10^{-12} on each of the four component sums separately. Branch choices and external citation provenance are recorded independently in L1 and L2.

For the Phase-1 prime-residue replication of § X.5.1, two analogous stacks **P1a** and **P1b** are used: **primesieve** 12.13 (segmented wheel sieve) and a hand-rolled plain-C segmented Eratosthenes sieve with no external dependency.

1.3. The local Perron double-pole residue. Lemma X.3.1. *Let χ be a primitive non-principal Dirichlet character and let ρ be a simple zero of $L(s, \chi)$. Then for any $K > 1$,*

$$(1) \quad \operatorname{Res}_{w=0} \left[\frac{K^w}{w L(w + \rho, \chi)} \right] = \frac{\log K}{L'(\rho, \chi)} + C_1(\chi, \rho).$$

Proof. Taylor-expand $L(w + \rho, \chi)$ at $w = 0$; invert to get $1/L(w + \rho, \chi) = (L'(\rho, \chi)w)^{-1} - L''(\rho, \chi)/(2L'(\rho, \chi)^2) + O(w)$. Multiply by $K^w/w = w^{-1} + \log K + O(w)$ and read off the coefficient of w^{-1} . $\square$

Lemma X.3.1 is unconditional given simplicity of ρ . The full algebraic derivation is in Appendix B § B.2; the identity is also machine-verified in Lean 4 / Mathlib v4.28.0 (`LocalPerronResidue.lean`, 0 sorry; see § X.6).

1.4. Identities.

1.4.1. *Corrected B_∞ identity (unconditional).* Our main algebraic result is the following unconditional identity for the prime-power tail T_∞ .

Theorem X.4.1. *Let χ be a primitive non-principal Dirichlet character of conductor q , let ρ be a simple zero of $L(s, \chi)$ on the critical line, and let ψ be the primitive character of conductor $f \mid q$ inducing χ^2 . Then*

$$(2) \quad T_\infty(\chi, \rho) = \frac{1}{2} \log L(2\rho, \psi) + \text{BPC}_1(\chi, \rho) + \text{BPC}_2(\chi, \rho) + T_{\geq 3}(\chi, \rho),$$

where

$$\begin{aligned} \text{BPC}_1 &= \frac{1}{2} \sum_{p \mid q, p \nmid f} \log(1 - \psi(p) p^{-2\rho}), \\ \text{BPC}_2 &= -\frac{1}{2} \sum_{k \geq 2} \frac{1}{k} \sum_p \frac{\chi(p)^{2k}}{p^{2k\rho}}, \quad T_{\geq 3} = \sum_{k \geq 3} \frac{1}{k} \sum_p \frac{\chi(p)^k}{p^{k\rho}}. \end{aligned}$$

The four right-hand-side terms are individually finite. The identity is unconditional given simplicity of ρ .

The $k = 1$ prime sum $\sum_p \chi^2(p)/p^{2\rho}$ is only conditionally convergent on $\operatorname{Re}(s) = 1$; its convergence is supplied by Akatsuka [Aka, Lemma 2.1 & eq. (2.5)], which is itself unconditional (derived from PNT with explicit error term). BPC_2 and $T_{\geq 3}$ are absolutely convergent (minimum exponents $\operatorname{Re}(2k\rho) = 2$ and $\operatorname{Re}(k\rho) = \frac{3}{2}$). The full proof is given in Appendix A.

1.4.2. *Subleading constant C_1 and the partial Möbius identity. Theorem X.4.2.* *Under the hypotheses of Theorem X.4.1, and assuming every off-target nontrivial zero ρ' of $L(s, \chi)$ with $|\gamma' - \tau| \leq T_K$ is simple (for a zero-avoiding height T_K as in the truncated explicit formula of Inoue [Ino, Theorem 1]),*

$$(3) \quad c_K(\chi, \rho) = \frac{\log K}{L'(\rho, \chi)} + C_1(\chi, \rho) + o(1) \quad (K \rightarrow \infty).$$

The identity is unconditional in ρ given off-target simplicity. The rate $o(1) = O(K^{-1/2+\epsilon})$ is RH-conditional; the unconditional Soundararajan [Sou] bound gives $o(1) = O(K^{-1/2} \exp((\log K)^{1/2} (\log \log K)^{14}))$.

The proof combines Inoue [Ino, Theorem 1]’s truncated explicit formula for $M^*(K, \chi)$ with Lemma X.3.1 to extract the double-pole residue at ρ ; off-target zero contributions are bounded by Soundararajan’s argument. See Appendix B for the full proof.

1.4.3. *The Aoki–Koyama–Mertens constant (Hypothesis AK).* Aoki–Koyama (2023, *J. Number Theory* **245**, eq. (1.4), p. 235) states for a non-principal Dirichlet χ that, under DRH,

$$\lim_{x \rightarrow \infty} \left((\log x)^m \prod_{p \leq x} (1 - \chi(p)/p^s)^{-1} \right) = \frac{L^{(m)}(s, \chi)}{e^{m\gamma} m!} \cdot \begin{cases} \sqrt{2}, & \chi^2 = 1, s = \frac{1}{2}, \\ 1, & \text{otherwise,} \end{cases}$$

with $m = m_\chi = \text{ord}_{s=1/2} L(s, \chi)$. Specialised to the regime of this paper ($m = 1$, $\rho \neq \frac{1}{2}$, branch multiplier 1), this reads:

$$(AK) \quad \lim_{K \rightarrow \infty} E_K(\chi, \rho) \log K = \frac{L'(\rho, \chi)}{e^\gamma}.$$

This **corrects** the earlier target $L'(\rho, \chi)/\zeta(2)$: the ratio is $\zeta(2)/e^\gamma \approx 1.0828$. (AK) is DRH-conditional in the form stated by Aoki–Koyama.

1.4.4. *The conditional NDC limit (open).* Composing (3) with (AK) formally gives

$$(NDC) \quad D_K(\chi, \rho) = c_K(\chi, \rho) E_K(\chi, \rho) \longrightarrow e^{-\gamma} \quad (K \rightarrow \infty),$$

the corrected Numerical Duality Constant. The composition is mechanical provided (3) is strengthened to the **shifted Perron leading theorem**:

$$(SP-L) \quad c_K(\chi, \rho) = \frac{\log K}{L'(\rho, \chi)} + o(\log K) \quad (K \rightarrow \infty).$$

The obstruction is the off-target nontrivial-zero residue aggregate: even under DRH, an off-target zero λ of multiplicity $m \geq 2$ contributes

$$K^{\lambda-\rho} (\log K)^{m-1} / ((m-1)! (\lambda-\rho) a_m(\lambda)),$$

and the $(\log K)^{m-1}$ factor is not absorbed by simplicity of ρ . A literature search ([Ino], [Sou], Ng [Ng2], [Aka]) did not surface a theorem that closes (SP-L); we state (NDC) as **conditional on (AK) and (SP-L)** (Question Q:Perron, § X.7).

1.5. Numerical findings.

1.5.1. *Phase-1 Dominance-of-−1 replication, $x = 1.3 \cdot 10^{13}$.* We independently replicate the prime-residue counts $\pi(x; N, a)$ of Koyama’s *nontriv.pdf* for $N \in \{7, 8, 11, 19, 23\}$ and $x \in \{10^{12}, 1.3 \cdot 10^{12}, 10^{13}, 1.3 \cdot 10^{13}\}$ via two independent prime-enumeration implementations (`primesieve` 12.13 and a hand-rolled segmented C sieve). Headline numbers:

- $\pi(1.3 \cdot 10^{13}) = 445,831,610,611$, cross-checked against `primesieve --count` standalone.
- Library-independence: at every one of the four checkpoints, the `primesieve` counts and the hand-rolled C-sieve counts agree on every residue class for every N .
- Hardware-independence: a second M1-class machine agrees through $x = 1.3 \cdot 10^{12}$ on every residue class for every N .
- Koyama’s identity (3.1), a Dirichlet-orthogonality cross-check on the residue-count vector, is verified directly at all 495 (N, x, a) -cells (worst absolute residual $1.4 \cdot 10^{-4}$).

Cell-by-cell comparison with Koyama’s Tables 3–7 at all four checkpoints, all moduli:

Table	N	cells	exact	substantive disagreement (at $x = 1.3 \cdot 10^{13}$)
3	7	12	11	1 cell ($\Delta = 50$, clean digit-shift profile)
4	8	12	1	11 (small- x rows; possible x -label error in Table 4 draft)
5	11	20	19	1 cell ($a = 10$: our 11,503 vs Koyama 71,711)
6	19	18	15	3 (2 substantive at $a = 13, 18$; 1 sign flip at small x)
7	23	30	29	1 cell ($\Delta = 100$, clean transposition profile)
Total		92	75	17 (74/81 91% excluding the 11 Table-4 small- x rows)

Comparing to the qualitative dominance-of- -1 statement of Koyama (*nontriv.pdf*, §3): the signal is **cleanly reproduced for** $N = 8$ ($-1 = 7$ is the strict largest at $1.3 \cdot 10^{13}$, $\pi(\cdot; 8, 7) - \pi(\cdot; 8, 1) = 164,958$ vs $126,732$ and $102,728$). For $N = 19$ the ranking of -1 at this checkpoint is 3rd of 9 non-residues in both our run and Koyama's table — consistent with -1 being in the top group but not strictly dominant. For $N = 11$ the dominance turns on the disputed cell $a = 10$ (Table 5; see below); with Koyama's reported 71,711, -1 ranks 2nd of 5 (top group); with our 11,503, -1 ranks 3rd of 5 (outside top group). Pending reconciliation of that cell. For $N = 7$ and $N = 23$, Koyama himself notes (*nontriv.pdf* p. 19) that the predicted bias is not yet cleanly observed at this scale and attributes it to the exceptionally low-lying first zero of the relevant L -functions; our data agrees (-1 is 2nd of 3 for $N = 7$, mid-rank for $N = 23$). Koyama's illustrative estimate (*nontriv.pdf* p. 19, for $N = 19$) places the next sine-wave peak of the leading complex character at $x = e^{33.4} \approx 3.2 \cdot 10^{14}$, indicating that strict dominance for the harder moduli is expected only at substantially larger x .

The full replication bundle (source code, build hashes, TSV outputs, and reproducibility manifest) is deposited as Supplementary Material S1.

1.5.2. *Numerical values of the four Dirichlet pairs.* The four pairs (χ, ρ) used throughout §X.5.2–§X.5.4 (all values computed in mpmath at 50 decimal places):

Pair	χ (conductor)	$\rho = \frac{1}{2} + i\tau$	$L'(\rho, \chi)$	$L''(\rho, \chi)$
χ_{-4}/z_1	χ_{-4} ($q = 4$)	$\frac{1}{2} + 6.020949i$	$1.296500 +$ $0.182765i$	$-1.697050 -$ $0.554017i$
χ_{-4}/z_2	χ_{-4}	$\frac{1}{2} + 10.243770i$	$1.788467 -$ $0.296776i$	$-3.319767 +$ $0.755548i$
χ_5	χ_5 ($q = 5$)	$\frac{1}{2} + 6.183578i$	$1.112930 -$ $0.448830i$	$-1.642973 +$ $1.035107i$

Pair	χ (conductor)	$\rho = \frac{1}{2} + i\tau$	$L'(\rho, \chi)$	$L''(\rho, \chi)$
χ_{11}	χ_{11} ($q = 11$)	$\frac{1}{2} + 3.547041i$	$1.696582 - 0.250988i$	$-3.121598 + 0.261219i$

L1b in-language cross-check (Hurwitz expansion, independent sieve) agrees with L1 to $|\Delta L'|, |\Delta L''| \lesssim 6 \cdot 10^{-12}$ and $|\Delta C_1| \lesssim 5 \cdot 10^{-13}$. L2 cross-language (PARI/GP 2.17.3) agrees with L1 to ≥ 11 decimal digits on every real and imaginary component. An Arb spot-check at 250 bits on the worst pair gives interval agreement on $|L'|$ within $3 \cdot 10^{-43}$.

1.5.3. *The Aoki–Koyama drift: $e^{-\gamma}$ vs $\zeta(2)^{-1}$. (Analytic-identity scale $K \leq 10^7$; not transferred from §X.5.1.)*

The modulus $|D_K|$ statistic at $K = 2 \cdot 10^6$ and $K = 10^7$ (40 dps, mean over the four pairs):

Quantity	$K = 2 \cdot 10^6$	$K = 10^7$	$\zeta(2)^{-1}$ target	$e^{-\gamma}$ target
Mean $ D_K \cdot \zeta(2)$	0.992	0.974	1.000	$\zeta(2) e^{-\gamma} \approx 0.9237$
Mean $ E_K \log K e^{\gamma} / L' $	n/a	0.942	n/a	1.000

The drift from 0.992 to 0.974 between $K = 2 \cdot 10^6$ and $K = 10^7$ is consistent with the AK normalisation $e^{-\gamma}$ at the natural $1/\log K$ finite-size scale and **incompatible with the $\zeta(2)^{-1}$ target** at the same scale. We do not claim convergence of the complex $D_K(\chi, \rho)$ from a modulus statistic alone — that depends on (SP-L), which is open.

1.5.4. *The B_∞ identity at the four pairs. (Analytic-identity scale $K \leq 10^8$; not transferred from §X.5.1.)*

Identity residual $|T_K - \text{RHS}|$ for (2) at three scales — $K = 2 \cdot 10^6$ (mpmath, 50 dps; cross-checked in PARI/GP 2.17.3, closed-form component evaluation), $K = 10^7$ and $K = 10^8$ (PARI/GP):

Pair	$K = 2 \cdot 10^6$ (L1)	$K = 2 \cdot 10^6$ (L2)	$K = 10^7$ (L2)	$K = 10^8$ (L2)
χ_{-4}/z_1	$2.85 \cdot 10^{-3}$	$2.85 \cdot 10^{-3}$	$2.58 \cdot 10^{-3}$	$2.25 \cdot 10^{-3}$
χ_{-4}/z_2	$1.66 \cdot 10^{-3}$	$1.66 \cdot 10^{-3}$	$1.52 \cdot 10^{-3}$	$1.32 \cdot 10^{-3}$
χ_5	$4.24 \cdot 10^{-5}$	$4.24 \cdot 10^{-5}$	$1.22 \cdot 10^{-5}$	$3.30 \cdot 10^{-6}$
χ_{11}	$3.34 \cdot 10^{-5}$	$3.34 \cdot 10^{-5}$	$1.75 \cdot 10^{-5}$	$4.10 \cdot 10^{-6}$

L1 and L2 agree to all displayed digits at $K = 2 \cdot 10^6$ (stack difference $\leq 10^{-8}$). The decay across three decades on the clean-character pairs χ_5, χ_{11} (where $\chi(2) \neq 0$ and there is no bad-prime contribution to BPC_1):

Pair	residual ratio $K = 2 \cdot 10^6 \rightarrow 10^7$	$K = 10^7 \rightarrow 10^8$	$\sqrt{5} \approx 2.24$	$\sqrt{10} \approx 3.16$
χ_5	3.5	3.7	2.24	3.16
χ_{11}	1.9	4.3	2.24	3.16

Both ratios bracket the predicted $K^{-1/2}$ behaviour from the boundary-line conditional tail (Akatsuka [Aka, eq. (2.5)]), well within the oscillatory $O(1)$ implicit-constant envelope of the Soundararajan-conditional rate. The χ_{-4} pairs show systematically slower decay (ratio ≈ 1.15 over each decade) consistent with the additional bad-prime $p = 2$ contribution to BPC_1 .

1.5.5. *Two negative elliptic-curve findings.* We record two negative findings on the elliptic-curve side, both deferred to the supplementary record for the design data, source, and resampling diagnostics:

- *Conductor-confounded rank trend.* A multivariate OLS regression of $E[C_1^2]$ on $(\text{rank}, \log N)$ across 19 weight-2 elliptic curves shows a stable $\log N$ coefficient and an *unstable* rank coefficient (bootstrap 95% CI for the rank coefficient includes zero; one rank-3 anchor swings the coefficient by 63%). We describe this as a conductor-confounded trend, not a rank law (Question Q:conductor, § X.7).
- *Raw $\text{Sym}^2/\langle f, f \rangle$ proportionality.* The raw ratio ranges over seven orders of magnitude across $\{37a_1, 389a_1, \Delta\}$ and is empirically falsified in its raw form (Question Q:Sym2, § X.7).

1.6. **Lean 4 / Mathlib4 formalisation.** We accompany the paper with a Lean 4 / Mathlib4 lake project (formal-conjectures/ directory of the supplementary archive). The toolchain is leanprover/lean4:v4.28.0; Mathlib is pinned at commit 8f9d9cff6bd728b17a24e163c9402775d9e6a365. The Lean inventory fixes the **statements** of every identity of § X.4, ensures normalisations and branch conventions are syntactically explicit, and records each statement’s proof status against a public audit trail.

Build status. lake build FormalConjectures succeeds on all **10 files** in formal-conjectures/ with **2 sorry warnings**, each annotated in-source as MATHLIB-PREREQ: or RESEARCH-OPEN:. **Eight files are fully proved (0 sorry):** all the algebraic content of § X.3, § X.4, the smoothed explicit-formula chain, the Mertens spectroscopy universality statement, the Farey bridge identity, and DPAC for $K \leq 4$. No axiom is introduced anywhere in the project. The full per-sorry inventory is in LEAN_SORRY_STATUS.md of the reproducibility bundle.

Status of headline theorems in § X.3–§ X.4:

Paper object	Lean status
Lemma X.3.1 (local Perron residue)	Lean-verified unconditional (LocalPerronResidue.lean, 0 sorry)
Theorem X.4.1 (corrected B_∞ identity)	Lean-verified conditional on the convergence-of-partial-sum hypothesis derived in Appendix A (CorrectedBInfy.lean, 0 sorry)
Theorem X.4.2 (c_K leading + subleading)	Pen-and-paper proof in Appendix B; off-target-zero-simplicity hypothesis stated explicitly
(AK), (SP-L), (NDC)	(AK) cited (Aoki–Koyama 2023); (SP-L) and (NDC) open challenges (Q:Perron, § X.7)
DPAC	Lean-verified for $K \leq 4$ unconditional (DPAC_closure_attempt.lean, 0 sorry); general K open (LI-class), with obstruction certificate via [Pol]

Paper object	Lean file	Status
Boundary residue $R_0 = -2$ for a Gaussian-cutoff Mellin-shift explicit formula (companion strand) and its algebraic-glue chain	<code>SmoothedDwfFormula_full.lean</code>	THEOREM (chain), 0 sorry. All 17 algebraic-glue lemmas closed unconditionally; the two analytic prerequisites <code>mellin_decay</code> (Stirling on Γ vertical strips) and <code>inv_zeta_polynomial_growth</code> (Titchmarsh [Tit, §3.11]) are now stated as explicit hypotheses on the theorems that consume them, both Mathlib v4.28.0 gaps.
Lemma X.3.1 (local Perron residue)	<code>LocalPerronResidue.lean</code>	THEOREM (0 sorry). The residue identity is stated as a <code>Tendsto</code> limit at L analytic with simple zero at 0 (the general- ρ case reduces by $L \mapsto L(\cdot + \rho)$).
Theorem X.4.1 (B_∞ identity)	<code>CorrectedBInfnty.lean</code>	THEOREM (0 sorry), conditional on <code>h_convergence</code>. The four-component identity is proved against <code>noncomputable defs</code> of T_∞ , $T_{\geq 3}$, BPC_1 , BPC_2 , L given an added hypothesis <code>h_convergence : Tendsto T_K atTop (nhds (RHS))</code> . This hypothesis packages exactly the four analytic inputs of the pen-and-paper proof in Appendix A ([Aka], log-Euler-product, imprimitive induction, geometric tails); the Lean proof uses <code>Classical.epsilon_spec + tendsto_nhds_unique</code> and is three lines.
Farey bridge identity	<code>FareyBridgeIdentity.lean</code>	THEOREM (0 sorry), conditional on a Ramanujan-sum decomposition hypothesis packaging the Hardy–Wright [Har, Theorem 304] input $(c_q(p) = \mu(q) \text{ for } \gcd(p, q) = 1)$. Two helper lemmas <code>(mertens_eq_pred_add_moebius, sum_moebius_Icc_eq_mertens)</code> closed without sorry. MATHLIB-PREREQ: <code>Mathlib.NumberTheory.RamanujanSum</code> for the unconditional version.

Paper object	Lean file	Status
Mertens spectroscopy universality	MertensSpectroscopyUniversality.lean	THEOREM (0 sorry), conditional on an explicit-formula asymptotic hypothesis (Soundararajan [Sou, Theorem 1] input). Proof: <code>Filter.tendsto_atTop.mpr h_explicit_formula</code> .
Farey sign pattern	FareySignPattern.lean	NEGATIVE. The pointwise version is falsified at $p = 237,733$ and $p = 243,799$ (recorded as theorems, not axioms — the project’s “no axiom” convention is preserved). The density-one surviving version is stated as a <code>Tendsto</code> . 3 sorrys .
Ramanujan sum at primes (Hardy & Wright Thm 304)	RamanujanSum.lean	THEOREM (0 sorry). For any $q \geq 1$ and prime p coprime to q , $c_q(p) = \mu(q)$. Uses the Möbius-inversion form $c_q(n) = \sum_{d \gcd(q,n)} d\mu(q/d)$; the proof reduces the divisor set to $\{1\}$ via gcd-commutativity. Discharges piece (1) of <code>FareyBridgeIdentity</code> ’s <code>h_ramanujan_decomp</code> ; full discharge still requires the Farey-sequence decomposition lemmas (Mathlib gap).

Paper object	Lean file	Status
Dirichlet Polynomial Avoidance (DPAC) — partial closure + bridges	DPAC_closure_attempt.lean, DirichletPolynomialAvoidance.lean DPAC_full.lean	PARTIAL. DPAC_closure_attempt.lean (0 sorry) proves DPAC unconditionally for $K \in \{2, 3, 4\}$ using only $0 < \text{Re}(\rho) < 1$ (dpac_K_eq_2, dpac_K_eq_3, dpac_K_eq_4, dpac_le_4). It also reformulates the open case as FiniteLogRatioLI and records the obstruction certificate ([Pol] discreteness of the exponential-polynomial zero set + a single open avoidance statement). The headline conjecture for general K remains sorry in DPAC_full.lean:297 and DirichletPolynomialAvoidance.lean:48, diagnostically comparable to the Linear Independence Hypothesis for ζ -zero ordinates; the four explicit phase-avoidance bridges (dpac_of_logPrimePhaseAvoidance, ..., dpac_of_certifiedZetaZeroSample) are closed without sorry.

The role of the Lean artifact is to fix the statements and provide a publicly inspectable audit trail of the proof obligations remaining.

1.7. Open challenges. The following structure the next phase of the program.

Q:Perron (Shifted Perron leading theorem). Prove (SP-L) ((SP-L)) for primitive non-principal χ and simple non-central ρ .

Three sufficient packages, in decreasing strength of input required (see SP_L_SUFFICIENT_PACKAGES_2026-05-13.md of the supplementary archive for the full discussion):

- **Route I (off-target simplicity + shifted negative second moment).** All off-target zeros at the truncation height are simple, and $\sum_{\rho'}^{\text{mult}} |L(\rho' + \alpha, \chi)|^{-2} \ll_{\chi} (\log K)^{O(1)}$. Near-Lindelöf strength; not unconditionally available.
- **Route II (halo-route reduction).** The cluster-summed-residue contour pivot (transferred from the GL(2) version) replaces the termwise budget by a signed cancellation but yields only $|R_K| \ll K^{1/2+\varepsilon}$ for the off-target aggregate — **far above the $o(\log K)$ target**. The halo route in its present form does not close (SP-L), though it does replace the rooted Palm wall as the structural obstruction.
- **Route III (direct partial summation).** Substantially weaker: a Gonek–Hejhal-type bound $A(T) := \sum_{\gamma' \leq T, \gamma' \neq \tau} 1/[(\rho' - \rho) |L'(\rho', \chi)|] \ll_{\chi} \log T$, combined with a Mertens-style oscillation bound $\int_0^T K^{i(\gamma-\tau)} dA(\gamma) = o(\log T)$ uniformly in K along the zero-avoiding heights. This is the cleanest pen-and-paper analogue of [Aka] eq.~(2.5) applied to the off-target weight sequence.

The total-Möbius bounds of Soundararajan type are too coarse to isolate the pointwise cancellation at the $\log K$ scale by themselves.

Q:EC-recipe (GL(2) reciprocal-derivative control). Prove a fixed-curve theorem for $\sum_{\gamma} \widehat{W}(i\gamma) e^{i\gamma u} / L'(E, 1 + i\gamma)$ giving cancellation $o(u^r)$, or a minimum-modulus estimate on a vertical line with explicit exponent < 2 .

Without a GL(2) analogue of Aoki–Koyama, the EC side remains at the level of quantitative ensemble evidence.

Q:DPAC (Dirichlet Polynomial Avoidance). Prove DPAC for general K . We give an unconditional Lean proof for $K \in \{2, 3, 4\}$; the general case reduces, via [Pol] discreteness of the finite-exponential-polynomial zero set, to a single open avoidance statement at ζ -zero ordinates diagnostically comparable to the Linear Independence Hypothesis.

Further questions (from the EC and ensemble negatives of § X.5.5; deferred to the supplementary record):

- *Q:conductor* — Replicate (the rank-vs-log N regression of § X.5.5) on a curve set in which rank and $\log N$ are not collinear, to separate the conductor contribution from any genuine rank dependence.
- *Q:Sym2* — Identify a completed / archimedean-corrected Sym^2 normalisation replacing the empirically falsified raw $\text{Sym}^2 / \langle f, f \rangle$ proportionality.
- *Q:EC-NDC* — Find a normalisation of D_K^E for which the universal limit exists and survives a null-control gate. The sharp-cutoff form $D_K^E \cdot \zeta(2) \rightarrow 1$ is falsified through $K = 10^6$; smoothed variants pass empirically but also pass a null-control gate against predeclared null transformations.

1.8. Code, data, and certificate availability. All scripts, refined zero data, numerical-table CSVs, convergence logs, the Lean 4 lake project, and the reproducibility manifest will be deposited as a single self-contained reproducibility bundle (Supplementary material S1), mirrored at a Zenodo DOI at acceptance. The bundle pins all software versions: Lean toolchain `leanprover/lean4:v4.28.0`, Mathlib commit `8f9d9cff6bd728b17a24e163c9402775d9e6a365`, `mpmath` 1.4, PARI/GP 2.17.3, FLINT 3.3 / `python-flint` 0.8.0. The Phase-1 Dominance-of-1 replication bundle is included as the Supplementary Material S1 archive. Each numerical table in § X.5 cites the L1 script and L2 reproducer; each external theorem cited in § X.4 has its PDF retrieval recipe, page/equation, and verbatim quote recorded in the citation audit (Supplementary S2, *Citation audit*).

APPENDIX A. FULL PROOF OF THEOREM X.4.1 (B_∞ IDENTITY)

This appendix gives the full proof of Theorem X.4.1 of the main text: for a primitive non-principal Dirichlet character χ of conductor q and a simple zero ρ of $L(s, \chi)$ on the critical line, with ψ denoting the primitive character of conductor $f \mid q$ that induces χ^2 ,

$$(\star) \quad T_\infty(\chi, \rho) = \frac{1}{2} \log L(2\rho, \psi) + \text{BPC}_1(\chi, \rho) + \text{BPC}_2(\chi, \rho) + T_{\geq 3}(\chi, \rho),$$

with the four terms on the right defined in § X.1 of the main text. The proof uses no hypothesis beyond simplicity of ρ as a zero of $L(s, \chi)$; in particular it does not use RH, GRH, EDRH, or DRH.

A.1. Setup — partial-sum decomposition. For a primitive non-principal Dirichlet character χ modulo $q \geq 2$ and a simple zero $\rho = \frac{1}{2} + i\tau$ with $\tau \neq 0$ of $L(s, \chi)$ on the critical line, define

$$T_K(\chi, \rho) := \sum_{p \leq K} \sum_{k \geq 2} \frac{\chi(p)^k}{k p^{k\rho}}.$$

Inside the partial sum, the order of summation may be swapped unconditionally because for each fixed prime $p \geq 2$,

$$\sum_{k \geq 2} \frac{(\chi(p)p^{-\rho})^k}{k} = -\log(1 - \chi(p)p^{-\rho}) - \chi(p)p^{-\rho},$$

with $|\chi(p)p^{-\rho}| = p^{-1/2} \leq 2^{-1/2} < 1$, so the inner series is absolutely convergent and the logarithm is taken along the principal branch with no ambiguity. Therefore we may re-index the partial sum by k first:

$$(4) \quad T_K(\chi, \rho) = \frac{1}{2} \sum_{p \leq K} \frac{\chi(p)^2}{p^{2\rho}} + \sum_{k \geq 3} \frac{1}{k} \sum_{p \leq K} \frac{\chi(p)^k}{p^{k\rho}}.$$

The only delicate object in passing to $K \rightarrow \infty$ is the $k = 2$ piece, since $\operatorname{Re}(2\rho) = 1$ lies on the boundary line where the Dirichlet prime sum $\sum_p \chi^2(p)p^{-s}$ is conditionally (not absolutely) convergent. We address this in § A.2. The $k \geq 3$ piece is absolutely convergent and is handled in § A.3.

A.2. Identification of the $k = 2$ partial sum.

A.2.1. *The squared character χ^2 and its primitive companion ψ .* For χ primitive of conductor q , define χ^2 by $\chi^2(n) := \chi(n)\chi(n)$ for $\gcd(n, q) = 1$ and $\chi^2(n) = 0$ otherwise. The character χ^2 is a (possibly imprimitive) Dirichlet character modulo q . By a standard textbook result (e.g. Montgomery–Vaughan, *Multiplicative Number Theory I*, Theorem 9.4 / Corollary 9.5), every Dirichlet character modulo q is induced from a unique primitive character ψ of some conductor $f \mid q$, in the sense that $\chi^2(n) = \psi(n)$ for $\gcd(n, q) = 1$ and $\chi^2(n) = 0$ for $\gcd(n, q) > 1$. The corresponding L -function identity is

$$(5) \quad L(s, \chi^2) = L(s, \psi) \cdot \prod_{p \mid q, p \nmid f} \left(1 - \frac{\psi(p)}{p^s}\right),$$

valid initially for $\operatorname{Re}(s) > 1$ and then by analytic continuation in the natural domain of the right-hand side.

A.2.2. *Log-Euler-product expansion of $\log L(s, \chi^2)$.* For $\operatorname{Re}(s) > 1$, taking the principal branch of the logarithm of the Euler product $L(s, \chi^2) = \prod_{p \nmid q} (1 - \chi^2(p)p^{-s})^{-1}$ gives

$$(6) \quad \log L(s, \chi^2) = \sum_{p \nmid q} \sum_{k \geq 1} \frac{\chi^2(p)^k}{k p^{ks}} = \sum_p \sum_{k \geq 1} \frac{\chi(p)^{2k}}{k p^{ks}},$$

where the second equality extends the prime sum to *all* primes (the contribution of $p \mid q$ vanishes because $\chi(p) = 0$ for those primes). The double sum is absolutely convergent in $\operatorname{Re}(s) > 1$.

Isolating the $k = 1$ term in (6) yields

$$(7) \quad \sum_p \frac{\chi(p)^2}{p^s} = \log L(s, \chi^2) - \sum_{k \geq 2} \frac{1}{k} \sum_p \frac{\chi(p)^{2k}}{p^{ks}}, \quad \operatorname{Re}(s) > 1.$$

A.2.3. *Analytic continuation of (7) to $s = 2\rho$.* We extend (7) to the half-plane $\operatorname{Re}(s) > \frac{1}{2}$ minus the singular set of the right-hand side, as follows.

Right-hand side analyticity at $s = 2\rho$. The function $\log L(s, \chi^2)$ is holomorphic at $s = 2\rho$ provided $L(2\rho, \chi^2) \neq 0$ and $L(s, \chi^2)$ has no pole there. When χ^2 is principal (i.e., $f = 1$, hence χ^2 is induced from the trivial character), $L(s, \psi) = \zeta(s)$, which has a simple pole at $s = 1$. But $s = 2\rho = 1 + 2i\tau$ with $\tau \neq 0$, so $s \neq 1$ and the pole of ζ is avoided. Non-vanishing of $L(s, \chi^2)$ on the line $\operatorname{Re}(s) = 1$ (away from the potential pole at $s = 1$) is the classical Hadamard–de la Vallée Poussin theorem (see e.g. Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*,

Chapter II.5). Hence $\log L(s, \chi^2)$ admits an analytic continuation through the neighbourhood of $s = 2\rho$ along the principal branch.

Right-hand $k \geq 2$ sum analyticity. For $k \geq 2$, $\operatorname{Re}(ks) \geq k \geq 2$ on the line $\operatorname{Re}(s) = 1$, so the sum $\sum_{k \geq 2} (1/k) \sum_p \chi(p)^{2k} p^{-ks}$ converges absolutely on $\operatorname{Re}(s) > 1/2$ and defines a holomorphic function there.

Left-hand side analyticity. The prime sum $\sum_p \chi^2(p) p^{-s}$ on the line $\operatorname{Re}(s) = 1$ (excluding $s = 1$ if χ^2 is principal) converges conditionally by a partial-summation argument due to Akatsuka [Aka], *The Euler product for the Riemann zeta function in the critical strip*, Lemma 2.1 and equation (2.5), which establishes that for $t_0 \neq 0$,

$$(8) \quad \sum_{p \leq X} \frac{1}{p^{1+2it_0}} = c(t_0) + O((\log X)^{-1}),$$

proved by partial summation against the prime number theorem with explicit error term. The estimate (8) is **unconditional** (it does not require RH or any GRH-type hypothesis); the same partial-summation argument applies with $\chi^2(p)/p^{1+2i\tau}$ in place of p^{-1-2it_0} , with the character orthogonality producing the appropriate cancellation. We denote the limit by $\Sigma_2(\chi, \rho) := \lim_{X \rightarrow \infty} \sum_{p \leq X} \chi^2(p) p^{-2\rho}$.

We pass to the boundary $s = 2\rho$ by Abel summation. For $\sigma > 1$, identity (7) reads $\log L(s, \chi^2) - \sum_{k \geq 2} (1/k) \sum_p \chi(p)^{2k} / p^{ks} = \sum_p \chi(p)^2 / p^s$, the left side analytic in a neighbourhood of $s = 2\rho$ (here we use Hadamard-de la Vallée Poussin: $L(s, \psi) \neq 0$ on $\operatorname{Re}(s) = 1$ with $s \neq 1$, where ψ is the primitive character inducing χ^2 , together with the imprimitive Euler-factor identity $L(s, \chi^2) = L(s, \psi) \prod_{p|q, p \nmid f} (1 - \psi(p)/p^s)$). Writing $S(X) := \sum_{p \leq X} \chi(p)^2 / p^{2\rho}$ and applying Abel summation gives, for $\sigma > 1$, $\sum_p \chi(p)^2 / p^s = (s - 2\rho) \int_2^\infty S(X) X^{-(s-2\rho)-1} dX + \lim_{X \rightarrow \infty} S(X) X^{-(s-2\rho)}$; Akatsuka [Aka, Lemma 2.1 & eq. (2.5)] provides $S(X) = c(\chi, \rho) + O(1/\log X)$ unconditionally as $X \rightarrow \infty$, so the limit term equals $c(\chi, \rho)$ at $s = 2\rho$ and the integral converges to a continuous function of s in a neighbourhood. Hence (7) extends by continuity to $s = 2\rho$ as the boundary-value identity

$$(9) \quad \Sigma_2(\chi, \rho) = \log L(2\rho, \chi^2) - \sum_{k \geq 2} \frac{1}{k} \sum_p \frac{\chi(p)^{2k}}{p^{2k\rho}}.$$

A.2.4. *Multiplying by 1/2.* Multiplying (9) by $\frac{1}{2}$ and rewriting $\log L(2\rho, \chi^2)$ via the imprimitive-induction identity (5) gives

$$(10) \quad \frac{1}{2} \Sigma_2(\chi, \rho) = \frac{1}{2} \log L(2\rho, \psi) + \frac{1}{2} \sum_{p|q, p \nmid f} \log(1 - \psi(p) p^{-2\rho}) - \frac{1}{2} \sum_{k \geq 2} \frac{1}{k} \sum_p \frac{\chi(p)^{2k}}{p^{2k\rho}}.$$

The bad-prime correction in the middle term is precisely the definition of BPC_1 from §X.1; the last term is BPC_2 .

A.3. Absolute convergence of the $k \geq 3$ tail. For each fixed prime p and each $k \geq 3$, $|\chi(p)^k p^{-k\rho}| \leq p^{-k/2}$, so the inner k -series is bounded by a geometric series:

$$\left| \sum_{k \geq 3} \frac{\chi(p)^k}{k p^{k\rho}} \right| \leq \sum_{k \geq 3} \frac{p^{-k/2}}{k} \leq \frac{p^{-3/2}}{3} \cdot \frac{1}{1 - p^{-1/2}}.$$

Summing over primes:

$$|T_{\geq 3}| \leq \frac{1}{3} \sum_p \frac{p^{-3/2}}{1 - p^{-1/2}} \leq \frac{1}{3(1 - 2^{-1/2})} \sum_p p^{-3/2} = \frac{P(3/2)}{3(1 - 2^{-1/2})} \approx 0.515,$$

where $P(3/2) = \sum_p p^{-3/2} \approx 0.45224$ is the prime zeta function at $s = 3/2$. The truncation tail is also estimable: by partial summation against PNT, for $\alpha > 1$, $\sum_{p>K} p^{-\alpha} \leq \frac{2K^{1-\alpha}}{(\alpha-1)\log K}$, so

$$|T_{\geq 3} - T_{\geq 3,K}| \leq \frac{2}{3(1-2^{-1/2})} \cdot \frac{K^{-1/2}}{\log K},$$

which is $\sim 3 \cdot 10^{-4}$ at $K = 10^6$ and $\sim 1 \cdot 10^{-4}$ at $K = 2 \cdot 10^6$.

A.4. Assembling the identity. Define

$$T_{\infty}(\chi, \rho) := \lim_{K \rightarrow \infty} T_K(\chi, \rho).$$

The limit exists by combining the two parts:

- The $k = 2$ part is $\frac{1}{2}\Sigma_2(\chi, \rho)$, whose limit exists by Akatsuka [Aka, Lemma 2.1 & eq. (2.5)] and equals the right-hand side of (10).
- The $k \geq 3$ part is absolutely convergent by § A.3.

Hence

$$T_{\infty}(\chi, \rho) = \frac{1}{2} \log L(2\rho, \psi) + \text{BPC}_1(\chi, \rho) + \text{BPC}_2(\chi, \rho) + T_{\geq 3}(\chi, \rho),$$

which is identity (★). □

A.5. Bad-prime correction for the four computational pairs. For the four (χ, ρ) pairs used in the numerical work of § X.5.4, the bad-prime correction BPC_1 takes the following character-explicit forms:

Character	q	χ^2 order	primitive ψ inducing χ^2	bad primes $f \quad \{p \mid q, p \nmid f\}$
χ_{-4}	4	1 (principal mod 4)	trivial character ($\Rightarrow L = \zeta$)	1 {2}
χ_5	5	2	Legendre symbol $(\cdot/5)$	5 $\emptyset$
χ_{11}	11	5	order-5 character mod 11	11 $\emptyset$

Explicitly: - For χ_{-4} : $\text{BPC}_1 = \frac{1}{2} \log(1 - 2^{-2\rho})$. For $\rho = \frac{1}{2} + i\tau$, this is $\frac{1}{2} \log(1 - 2^{-1-2i\tau})$, of modulus $\leq \frac{1}{2} \log(1/(1 - 1/2)) = \frac{1}{2} \log 2 \approx 0.347$. - For χ_5 and χ_{11} : $\text{BPC}_1 = 0$.

This explains the numerical observation in § X.5.4 that the χ_{-4} pairs show slower convergence of the B_{∞} identity residual than χ_5 and χ_{11} : the bad-prime contribution from $p = 2$ adds an extra layer of conditionally-convergent boundary mass to the χ_{-4} identity that is absent for χ_5, χ_{11} (the latter pure conditional-tail decay scales as $K^{-1/2}/\log K$ exactly).

A.6. What the proof does *not* use. For clarity:

- The proof does **not** use RH, GRH, EDRH, or DRH. The only conditional convergence on the boundary line $\text{Re}(s) = 1$ is the $k = 1$ prime sum $\sum_p \chi^2(p)p^{-2\rho}$, handled by Akatsuka [Aka, Lemma 2.1 & eq. (2.5)], which itself is *unconditional* — derived from PNT with explicit error term.
- The proof does **not** use any rate of convergence for the partial Möbius / spectroscopy sum $c_K(\chi, \rho)$. The B_{∞} identity is a statement about the limit T_{∞} ; it makes no claim about the *speed* at which $T_K \rightarrow T_{\infty}$.

- The proof does **not** establish the Aoki–Koyama–Mertens limit ((AK)) for $E_K \log K$, which is a separate statement on the *additive* side and is cited as Hypothesis AK (with verbatim quotation in §X.4.3).
- The proof does **not** establish the conditional NDC limit ((NDC)); for that, Hypothesis AK and the shifted Perron leading hypothesis (SP-L) are both needed (see §X.4.4 and §X.7).

A.7. Lean formalisation. The four-component identity is formalised in Lean 4 / Mathlib v4.28.0 as the theorem `corrected_B_infty` in `formal-conjectures/CorrectedBInfty.lean` of the supplementary archive. The Lean proof is parameterised by an explicit `Filter.Tendsto` hypothesis asserting that the partial-sum sequence $T_K(\chi, \rho)$ converges to the four-component right-hand side as $K \rightarrow \infty$. **The convergence-as-hypothesis is precisely the conclusion of the present appendix (7) + (9) + the $k \geq 3$ tail of §A.3.** Given the convergence, the Lean proof is three lines: `unfold T_inf` (the `Classical.epsilon` of the `Tendsto` predicate), `Classical.epsilon_spec` (which yields that T_∞ inherits the same limit), and `tendsto_nhds_unique` (since $\mathbb{C}$ is Hausdorff). The file uses only the standard `propext`, `Classical.choice`, `Quot.sound` axioms.

A fully unconditional Lean proof of `corrected_B_infty` (i.e., one that *derives* the convergence rather than taking it as a hypothesis) requires upstream Mathlib formalisations of [Aka] eq. (2.5) and the imprimitive-induction Euler-factor identity for $L(s, \chi^2)$; both are MATHLIB-PREREQ and not yet upstream as of v4.28.0.

A.8. Numerical verification (summary; full table in §X.5.4). The identity $(\star)$ was verified numerically at $K = 2 \cdot 10^6$, 50 decimal places of precision, on the four (χ, ρ) pairs of §X.5.2. The residuals are:

Pair	residual $ T_K - \text{RHS}(\star) $ at $K = 2 \cdot 10^6$
χ_{-4}/z_1	$2.85 \cdot 10^{-3}$
χ_{-4}/z_2	$1.66 \cdot 10^{-3}$
χ_5	$4.24 \cdot 10^{-5}$
χ_{11}	$3.33 \cdot 10^{-5}$

These match the predicted $K^{-1/2}/\log K$ decay envelope at the $O(1)$ implicit-constant level (with the χ_{-4} pairs showing the predicted slower envelope due to the $p = 2$ bad-prime weight; see §A.5). The full table with per-component breakdown is at §X.5.4 of the main text.

APPENDIX B. FULL PROOF OF THEOREM X.4.2 (c_K LEADING + SUBLEADING IDENTITY)

This appendix gives the full proof of Theorem X.4.2 of the main text: under the hypotheses of Theorem X.4.1 (i.e., χ primitive non-principal of conductor q , $\rho = \frac{1}{2} + i\tau$ a *simple* zero of $L(s, \chi)$ with $\tau \neq 0$), and assuming every off-target nontrivial zero $\rho' \neq \rho$ of $L(s, \chi)$ in the truncation height $|\gamma'| \leq T_K$ of the Inoue [Ino] explicit-formula box is also *simple*,

$$(\dagger) \quad c_K(\chi, \rho) = \frac{\log K}{L'(\rho, \chi)} + C_1(\chi, \rho) + o(1) \quad (K \rightarrow \infty),$$

where $C_1(\chi, \rho) = -L''(\rho, \chi)/(2L'(\rho, \chi)^2)$ and $c_K(\chi, \rho) = \sum_{n \leq K} \mu(n) \chi(n) n^{-\rho}$.

The identity $(\dagger)$ is **unconditional in ρ** given simplicity of ρ and of the crossed off-target zeros. The $o(1)$ rate $O(K^{-1/2+\varepsilon})$ for every $\varepsilon > 0$ requires RH for $L(s, \chi)$; the *unconditional* rate (which still yields $o(1)$) is $O(K^{-1/2} \exp((\log K)^{1/2} (\log \log K)^{14}))$ via Soundararajan [Sou, Theorem 1].

B.1. Setup — Perron formula for c_K via $1/L(s, \chi)$.

B.1.1. *The Dirichlet series for $1/L(s, \chi)$.* For χ primitive of conductor $q \geq 2$ and $\text{Re}(s) > 1$,

$$\frac{1}{L(s, \chi)} = \sum_{n=1}^{\infty} \frac{\mu(n) \chi(n)}{n^s},$$

absolutely convergent on $\text{Re}(s) > 1$. The Dirichlet coefficients are $a_n = \mu(n) \chi(n)$, and the partial sum $c_K(\rho, \chi) = \sum_{n \leq K} a_n n^{-\rho}$ is the spectroscopy object.

B.1.2. *Perron formula (truncated).* The standard truncated Perron formula (Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, Theorem II.2.3; Inoue (2021), *Some explicit formulas for partial sums of Möbius functions*, Journal de Théorie des Nombres de Bordeaux **33** (2021), 273–315, eq. (4.1) and Theorem 1) gives the following representation.

For any $\sigma'_0 > \frac{1}{2}$ and $T \geq 2$, with the convention that c_K uses the half-weight at integer K ,

$$(11) \quad c_K(\rho, \chi) = \frac{1}{2\pi i} \int_{\sigma'_0 - iT}^{\sigma'_0 + iT} \frac{K^w}{w} \cdot \frac{1}{L(w + \rho, \chi)} dw + \mathcal{R}'(K, T, \rho),$$

where the kernel is K^w/w and the Dirichlet series factor is $1/L(w + \rho, \chi)$, obtained by the substitution $s = w + \rho$. The truncation error $\mathcal{R}'(K, T, \rho)$ is bounded by Inoue (2021) Theorem 1 via the J_1, J_2, J_3 estimates of that paper:

$$\mathcal{R}'(K, T, \rho) \ll \frac{K^{\sigma'_0 - 1/2} \log K}{T} + \min\left(1, \frac{K^{1/2}}{T \langle K \rangle}\right),$$

with $\langle K \rangle$ the distance from K to the nearest integer.

B.1.3. *Inoue [Ino, Theorem 1] — truncated explicit formula.* The following statement is verbatim from Inoue [Ino], arXiv:1805.05015v1, page 3:

Theorem 1 ([Ino]). Let $x > 0$, $q \geq 2$, $T \geq \max\{T_0, \exp(q^{1/3}), 2/x\}$. Then, uniformly for all primitive Dirichlet characters χ modulo d with $d \leq q$, there exists $T_\nu \in [T, 2T]$ satisfying

$$M^*(x, \chi) = \sum_{|\gamma| < T_\nu} \frac{1}{(m(\rho) - 1)!} \lim_{s \rightarrow \rho} \frac{d^{m(\rho)-1}}{ds^{m(\rho)-1}} \left((s - \rho)^{m(\rho)} \frac{x^s}{L(s, \chi) s} \right) + \text{Res}_{s=0} \left(\frac{x^s}{L(s, \chi) s} \right) + \dots$$

This is the unshifted form, applied to the partial-character-Möbius sum $M^*(x, \chi) = \sum_{n \leq x} \mu(n) \chi(n)$. Dividing by K^ρ and specializing at $x = K$, $m(\rho) = 1$ (the simplicity hypothesis on ρ), and after the change of variable $w = s - \rho$, one obtains (11).

B.2. Pole-structure analysis at $w = 0$. The pole-structure computation of this section is independently formalised in Lean 4 / Mathlib v4.28.0 as `local_perron_residue` in `formal-conjectures/LocalPerronResidue.lean`. The Lean theorem states the residue identity as a `Tendsto` limit at the canonical form L analytic with a simple zero at 0 (the general- ρ case of this appendix reduces to the canonical form by the substitution $L \mapsto L(\cdot + \rho)$); the proof is **unconditional** (0 sorry, only `propext`, `Classical.choice`, `Quot.sound` axioms).

We work in the shifted variable $w = s - \rho$. Define

$$F(w) := \frac{K^w}{w \cdot L(w + \rho, \chi)}.$$

Since $L(w + \rho, \chi)$ has a *simple* zero at $w = 0$ (by hypothesis), $1/L(w + \rho, \chi)$ has a simple pole at $w = 0$. Combined with the simple pole of K^w/w at $w = 0$, the integrand F has a **double pole** at $w = 0$.

B.2.1. *Laurent expansion of $1/L(w + \rho, \chi)$.* By Taylor expansion at the simple zero,

$$L(w + \rho, \chi) = a_1 w + a_2 w^2 + a_3 w^3 + \cdots, \quad a_k = \frac{L^{(k)}(\rho, \chi)}{k!}, \quad a_1 = L'(\rho, \chi) \neq 0.$$

Therefore

$$\frac{1}{L(w + \rho, \chi)} = \frac{1}{a_1 w} \cdot \frac{1}{1 + (a_2/a_1)w + (a_3/a_1)w^2 + \cdots}.$$

Expanding the geometric factor as $(1 + u)^{-1} = 1 - u + u^2 - \cdots$ and substituting $a_1 = L'(\rho, \chi)$, $a_2 = L''(\rho, \chi)/2$:

$$(12) \quad \frac{1}{L(w + \rho, \chi)} = \frac{1}{L'(\rho, \chi) w} - \frac{L''(\rho, \chi)}{2 L'(\rho, \chi)^2} + O(w).$$

B.2.2. *Laurent expansion of K^w/w .* Trivially,

$$\frac{K^w}{w} = \frac{1}{w} (1 + (\log K) w + \frac{1}{2} (\log K)^2 w^2 + O(w^3)) = \frac{1}{w} + \log K + \frac{1}{2} (\log K)^2 w + O(w^2).$$

B.2.3. *Product expansion and the double-pole residue.* Multiplying the two expansions:

$$F(w) = \left[\frac{1}{w} + \log K + \frac{1}{2} (\log K)^2 w + \cdots \right] \cdot \left[\frac{1}{L'(\rho, \chi) w} - \frac{L''(\rho, \chi)}{2 L'(\rho, \chi)^2} + O(w) \right].$$

Collecting powers of w : $-w^{-2}$ coefficient: $\frac{1}{L'(\rho, \chi)}$ (from $\frac{1}{w} \cdot \frac{1}{L'(\rho, \chi) w}$); $-w^{-1}$ coefficient: $\frac{\log K}{L'(\rho, \chi)} - \frac{L''(\rho, \chi)}{2 L'(\rho, \chi)^2}$ (cross-terms).

Hence

$$(13) \quad \text{Res}_{w=0} F(w) = \frac{\log K}{L'(\rho, \chi)} - \frac{L''(\rho, \chi)}{2 L'(\rho, \chi)^2} = \frac{\log K}{L'(\rho, \chi)} + C_1(\chi, \rho).$$

This is **Lemma X.3.1 of the main text**, established algebraically; the local part of ($\dagger$) is now done. It remains to control the off-target zero residues, the trivial-zero residues, and the contour pieces.

B.3. Off-target zero residues, trivial-zero residues, and contour pieces. We now shift the Perron contour from $\text{Re}(w) = \sigma'_0 > \frac{1}{2}$ leftward to a zero-avoiding line $\text{Re}(w) = -A$ for some $A > 0$, crossing all the nontrivial zeros of $L(s, \chi)$ in the strip $-A \leq \text{Re}(w) \leq \sigma'_0$. The integrand $F(w)$ has poles at $w = 0$ (the target double pole, with residue (13)), at $w = \rho' - \rho$ for every other nontrivial zero $\rho' \neq \rho$ of $L(s, \chi)$, at trivial zeros, and at $w = -\rho$ from a (potential) pole of $L(s, \chi)$ at $s = 0$ — but $L(s, \chi)$ is entire for primitive non-principal χ , so the pole at $w = -\rho$ from $1/L$ is absent. We do pick up the term $\text{Res}_{s=0}(x^s/(L(s, \chi)s))$ from Inoue's formula (this is the $s = 0$ residue in the unshifted form; after dividing by K^ρ it contributes a term of magnitude $K^{-\rho}/(L(0, \chi)) = O(K^{-1/2})$ in modulus, $o(1)$).

B.3.1. *Off-target nontrivial-zero residues.* For each $\rho' \neq \rho$ a nontrivial zero of $L(s, \chi)$, write $w_{\rho'} = \rho' - \rho$. Assuming ρ' is *simple* (which is generically expected and is the regime relevant for the leading + subleading identity), the residue of F at $w = w_{\rho'}$ is

$$\text{Res}_{w=w_{\rho'}} F(w) = \frac{K^{w_{\rho'}}}{w_{\rho'} \cdot L'(\rho', \chi)} = \frac{K^{\rho' - \rho}}{(\rho' - \rho) L'(\rho', \chi)}.$$

Under RH for $L(s, \chi)$, $|K^{\rho' - \rho}| = 1$ for all $\rho' = \frac{1}{2} + i\gamma'$ with $\gamma' \in \mathbb{R}$. The aggregate off-target sum (over zeros ρ' in the strip $|\gamma'| \leq T$) is then bounded by

$$\left| \sum_{\rho' \neq \rho, |\gamma'| \leq T} \frac{K^{i(\gamma' - \tau)}}{(\rho' - \rho) L'(\rho', \chi)} \right| \ll \sum_{\rho' \neq \rho, |\gamma'| \leq T} \frac{1}{|\gamma' - \tau| |L'(\rho', \chi)|}.$$

This is the off-target *aggregate* whose control determines the $o(1)$ rate.

B.3.2. Trivial-zero residues. Trivial zeros of $L(s, \chi)$ are at $s = -2k$ (χ even) or $s = -2k - 1$ (χ odd), $k \geq 0$. After the shift $w = s - \rho$, these contribute residues of size $K^{-\operatorname{Re}(\rho)-2k} = K^{-1/2-2k}$, summable to $O(K^{-1/2})$, which is $o(1)$.

B.3.3. Shifted contour pieces. The shifted contour $\operatorname{Re}(w) = -A$ contributes, on a suitable zero-avoiding height $T_K \in [K/(\log K)^B, 2K/(\log K)^B]$, an integral bounded by (using a reciprocal- L bound on the vertical shifted line)

$$\left| \frac{1}{2\pi i} \int_{(-A)} \frac{K^w}{w L(w + \rho, \chi)} dw \right| \ll K^{-A} \cdot T_K \cdot \exp(C(\log \log T_K)^2).$$

For $A > 1/2$ and $T_K \leq K/(\log K)^B$ with B sufficiently large, this is $K^{-A+1+o(1)} \cdot \exp(O((\log \log K)^2))$, which is $o(1)$.

Horizontal pieces give factors like $(T_K/K)^A \cdot \text{polylog}$, $o(1)$ for $T_K \leq K(\log K)^{-B}$ and large enough B .

B.3.4. The Perron truncation error. By Inoue [Ino, Theorem 1], the truncation error $\mathcal{R}'(K, T_K, \rho)$ is bounded by

$$\mathcal{R}'(K, T_K, \rho) \ll \frac{K^{\sigma'_0-1/2} \log K}{T_K} + \min\left(1, \frac{K^{1/2}}{T_K \langle K \rangle}\right).$$

For $T_K \asymp K(\log K)^{-B}$ and $\sigma'_0 = 1/2 + 1/\log K$, both pieces are $O(K^{-1/2+\varepsilon}) \cdot (\log K)^{O(1)}$ for any $\varepsilon > 0$, in particular $o(1)$.

B.3.5. Assembly. Combining the local residue (13) with the off-target and contour pieces:

$$c_K(\chi, \rho) = \frac{\log K}{L'(\rho, \chi)} + C_1(\chi, \rho) + (\text{off-target aggregate}) + O(K^{-1/2}) + O(K^{-A+1+o(1)} \exp(O((\log \log K)^2))) + \mathcal{R}$$

All terms in the last three parentheses are $o(1)$. The off-target aggregate is also $o(1)$ under RH for $L(s, \chi)$, by partial summation against Soundararajan [Sou], *Partial sums of the Möbius function*, Annals of Mathematics **170** (2009), 981–993, Theorem 1:

$$([\text{Sou}]) \quad \text{under RH: } M(x) \ll \sqrt{x} \exp((\log x)^{1/2} (\log \log x)^{14}).$$

Translating to $M^*(x, \chi)$ (the character analogue) gives the required $o(1)$ rate for the off-target aggregate under RH for $L(s, \chi)$.

This establishes (†). □

B.4. Rate of the $o(1)$. The proof above yields the following rates.

Rate	Hypotheses
$o(1) = O(K^{-1/2+\varepsilon})$ for every $\varepsilon > 0$	RH for $L(s, \chi)$.
$o(1) = O(K^{-1/2} \exp((\log K)^{1/2} (\log \log K)^{14}))$	unconditional ([Sou] applies under RH for $L(s, \chi)$, which holds for the four characters $\chi_{-4}, \chi_5, \chi_{11}, \dots$ at the verified heights).
$o(1) = O(\log K / \sqrt{K})$	RH plus a Gonek–Hejhal-type bound of the form $\sum_{0 < \gamma < T} L'(\rho, \chi) ^{-2} \ll T(\log T)^{O(1)}$.

In the numerical work of § X.5.3, the empirical residual $R(K) := c_K - \log K / L'(\rho, \chi) - C_1(\chi, \rho)$ at $K = 2 \cdot 10^6$ falls in the interval $[0.027, 0.375]$ across the four pairs; this is consistent with the Soundararajan-conditional rate within an $O(1)$ implicit-constant factor (specifically, $|R(K)|/(\log K / \sqrt{K}) \in [3, 36]$, well within the typical $\exp(C(\log \log K)^{1/2} (\log \log K)^{14})$ envelope at this scale).

B.5. What the proof does *not* claim. For clarity:

- The proof does **not** establish the Aoki–Koyama–Mertens limit for $E_K \log K$, which is the *multiplicative-side* asymptotic.
- The proof does **not** establish the *full* Perron-leading theorem $c_K = \log K/L'(\rho) + o(\log K)$ unconditionally; the off-target aggregate’s $o(\log K)$ control (which is *weaker* than the $o(1)$ obtained in (†) at the *leading + subleading* level) depends on RH for $L(s, \chi)$ and is what is required for composing with (AK) to obtain (NDC).
- The proof does **not** address the possibility of off-target *multiple* zeros, which under DRH/EDRH alone are not excluded; the manuscript’s open Question Q:Perron asks for either the off-target multiple-zero residue control or a sufficient “all crossed off-target zeros simple” hypothesis.

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